

HW 1

19. $d = d = 24$ -vector $w = 68$ -vector.

$$\vec{w} = \begin{bmatrix} d \\ d \\ d \\ d \\ d \\ d \\ d \end{bmatrix}$$

1b. $\vec{d} = \vec{X} \times \vec{W}$ cross product $\vec{X} = [I_{24} \ 0_{24 \times 144}]$

✓ 24×1 24×168 168×1

	$n=1$	$n=2$	$n=24$	$n=25$
$n=1$	1	0	0	0
$n=2$	0	1		

~~10~~ =

$n=24$ 0 . . . 0 . . . 0

- 2a. For most products and services the customer does not spend any dollars annually.
 b. For most stocks the dollar value holdings of the portfolio is worth nothing.
 c. For most needed materials the bill of materials are not needed.
 d. For most pixels the brightness is zero.
 e. For most days the location does not receive rainfall.

3. Total profit = $\vec{p}^T \cdot \vec{z}$

4a. $\text{Sym} = 1^T \vec{z}$

4b. $\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 0 \end{bmatrix}^T \vec{z}$

Any five 1's for $n=1$ to $n=10$
 0 for $n=11$ to $n=20$

5. $C = (1, 0, \dots, 0, -(1+r)^{T-1})$

$l_1 = (1, -(1+r), 0, 0)$

$l_2 = (0, 1, -(1+r), 0)$

$l_3 = (0, 0, 1, -(1+r))$

Pattern $\begin{bmatrix} 1 \\ 0 \\ 0 \\ -(1+r)^3 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ -(1+r) \\ 0 \\ 0 \end{bmatrix} + (1+r) \begin{bmatrix} 0 \\ 1 \\ -(1+r) \\ 0 \end{bmatrix} + (1+r)^2 \begin{bmatrix} 0 \\ 0 \\ 1 \\ -(1+r) \end{bmatrix}$

$C = l_1 + (1+r)l_2 + (1+r)^2l_3 + \dots + (1+r)^{T-2}l_{T-1}$

$$\begin{aligned}
 6a. (a+b)^T (a-b) &= \|a\|^2 - \|b\|^2 \\
 &= a^T a + b^T a - a^T b - b^T b \quad \text{since } b^T a = a^T b \\
 &= a^T a - b^T b \quad \text{since } \|a\| = \sqrt{a^T a} \\
 &= \|a\|^2 - \|b\|^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 6b. \|a+b\|^2 + \|a-b\|^2 &= 2(\|a\|^2 + \|b\|^2) \\
 &= \|a\|^2 + 2a^T b + \|b\|^2 + \|a\|^2 - 2a^T b + \|b\|^2 \\
 &= \|a\|^2 + \|b\|^2 + \|a\|^2 + \|b\|^2 \\
 &= 2\|a\|^2 + 2\|b\|^2 \\
 &= 2(\|a\|^2 + \|b\|^2) \quad \checkmark
 \end{aligned}$$

Code and Answers:

```
from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Part a

```
wine = load_wine()
X = wine.data
feature_names = wine.feature_names
print("X.shape:", X.shape)
print("Number of features:", X.shape[1])
print("Feature names:", feature_names)
X[:5]
```

```
df = pd.DataFrame(X)
df.head()
```

#The features represent the numeric measurements for wine samples. The features include malic acid, ash, alkalinity of ash, magnesium, total phenols, flavanoids, nonflavanoid phenol, proanthocyanins,

#color intensity, hue od280/od315 of diluted wines, and proline.

#Part b

```
scaler = StandardScaler()
Z = scaler.fit_transform(X)
print("Means of Z:", np.mean(Z, axis=0))
print("Stds of Z:", np.std(Z, axis=0))
```

#All the means are very close to 0, and the standard deviations are all equivalent to 1.

#Part c

```
inertias = []
for k in range(2, 9):
    kmeans = KMeans(n_clusters=k, random_state=0, n_init=10)
    kmeans.fit(Z)
    labels = kmeans.labels_
    centroids = kmeans.cluster_centers_
    plt.figure(figsize=(6, 4))
    plt.scatter(Z[:, 0], Z[:, 1], c=labels, cmap='viridis', s=30)
    plt.scatter(centroids[:, 0], centroids[:, 1], marker = "x")
    plt.title(f"K-Means Clustering on Wine Dataset (k = {k})")
    plt.xlabel("Standardized Alcohol")
    plt.ylabel("Standardized Malic Acid")
    plt.show()
```

#Based on the charts of the clustering between the standardized alcohol and malic acid, the k that provides the best distinguished groups is when $k = 3$.
 #The following k values do slightly improve, but they seem to overlap each other more, and it is honestly difficult to tell where the split occurs. However,
 #for $k = 3$, the groups are clearly shown to be on the left, right, and between with very slight overlap. This is also shown by the centroids.
 #The difference from $k = 2$ to $k = 3$ is very noticeable, while this does not really improve with the higher values of k. The higher values show the centroids being very close
 #to one another, and not really making distinguished groups.

Output:

X.shape: (178, 13)

Number of features: 13

Feature names: ['alcohol', 'malic_acid', 'ash', 'alcalihity_of_ash', 'magnesium', 'total_phenols', 'flavanoids', 'nonflavanoid_phenols', 'proanthocyanins', 'color_intensity', 'hue', 'od280/od315_of_diluted_wines', 'proline']

Means of Z: [7.84141790e-15 2.44498554e-16 -4.05917497e-15 -7.11041712e-17
 -2.49488320e-17 -1.95536471e-16 9.44313292e-16 -4.17892936e-16
 -1.54059038e-15 -4.12903170e-16 1.39838203e-15 2.12688793e-15
 -6.98567296e-17]

Stds of Z: [1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1. 1.]

